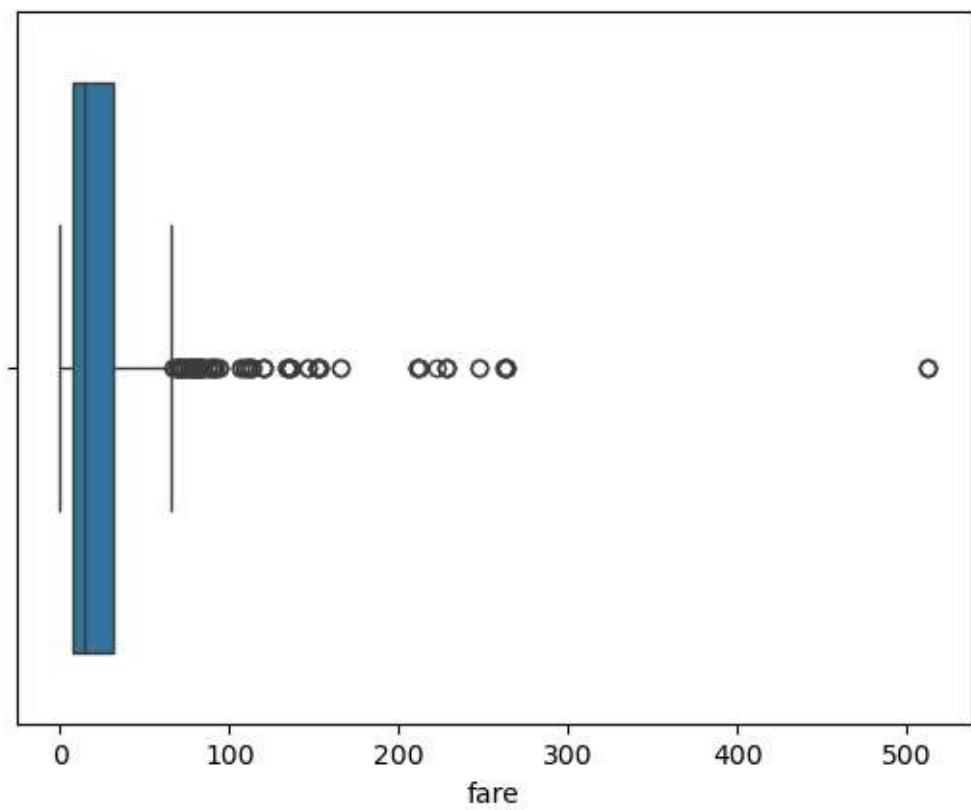


```
In [30]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import zscore
df = sns.load_dataset('titanic')
sns.boxplot(x=df['fare'])
plt.show()
```



```
In [31]: import pandas as pd
import seaborn as sns
from scipy.stats import zscore
df = sns.load_dataset('titanic')
Q1 = df['fare'].quantile(0.25)
Q3 = df['fare'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers_iqr = df[(df['fare'] < lower) | (df['fare'] > upper)]
print("IQR Outliers:", len(outliers_iqr))
df['fare_zscore'] = zscore(df['fare'].dropna())
df['fare_zscore'] = pd.Series(zscore(df['fare'].dropna()),
index=df['fare'].dropna().index)
outliers_z = df[abs(df['fare_zscore']) > 3]
print("Z-score Outliers:", len(outliers_z))
```

IQR Outliers: 116

Z-score Outliers: 20

```
In [32]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset('titanic')
print(df.head())
print(df.info())
print(df.columns)
print(df['fare'].head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked
0	0	3	male	22.0	1	0	7.2500	S
1	1	1	female	38.0	1	0	71.2833	C
2	1	3	female	26.0	0	0	7.9250	S
3	1	1	female	35.0	1	0	53.1000	S
4	0	3	male	35.0	0	0	8.0500	S

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	survived	891 non-null	int64
1	pclass	891 non-null	int64
2	sex	891 non-null	object
3	age	714 non-null	float64
4	sibsp	891 non-null	int64
5	parch	891 non-null	int64
6	fare	891 non-null	float64
7	embarked	889 non-null	object
8	class	891 non-null	category
9	who	891 non-null	object
10	adult_male	891 non-null	bool
11	deck	203 non-null	category
12	embark_town	889 non-null	object
13	alive	891 non-null	object
14	alone	891 non-null	bool

dtypes: bool(2), category(2), float64(2), int64(4), object(5)

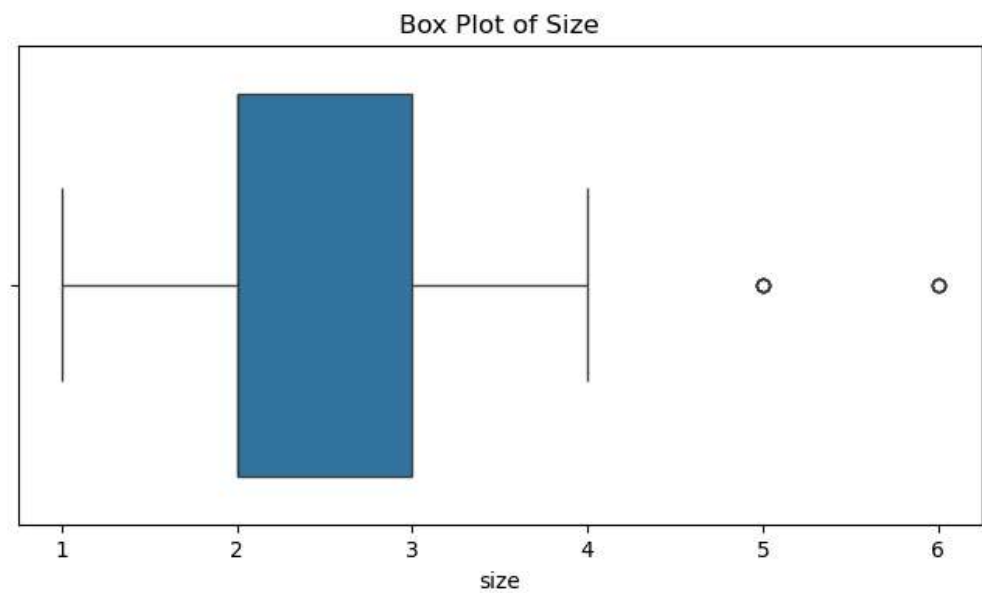
memory usage: 80.7+ KB

None

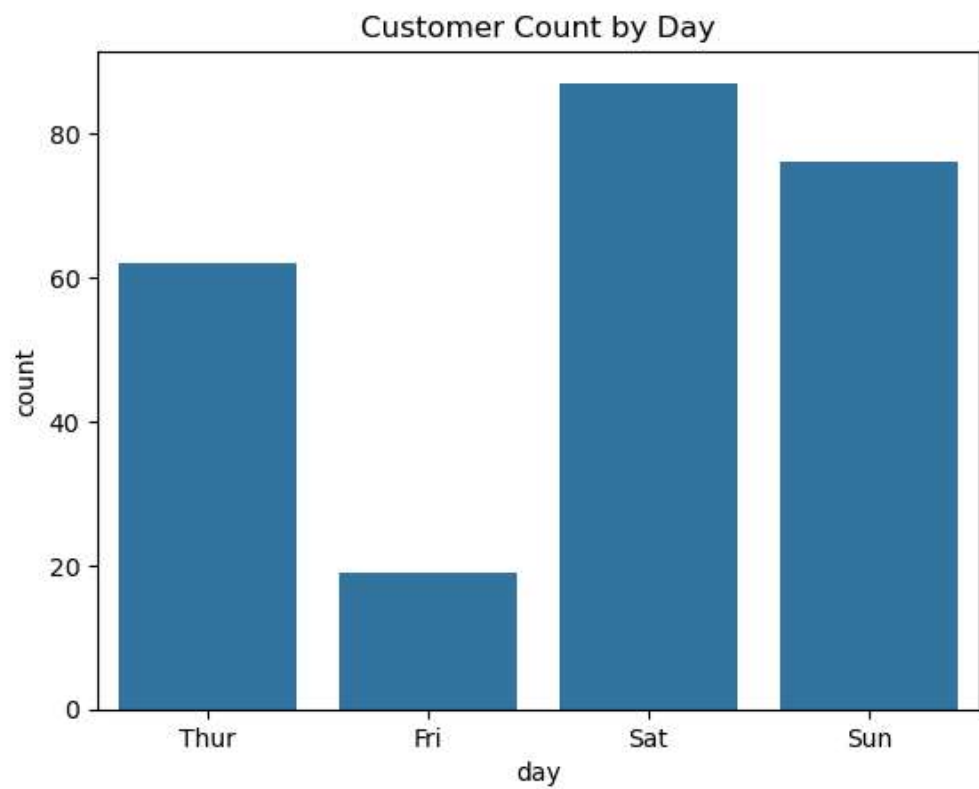
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
'embarked', 'class', 'who', 'adult_male', 'deck',

```
'embark_town',  
    'alive', 'alone'],  
    dtype='object')  
0    7.2500  
1   71.2833  
2    7.9250  
3   53.1000  
4    8.0500  
Name: fare, dtype: float64
```

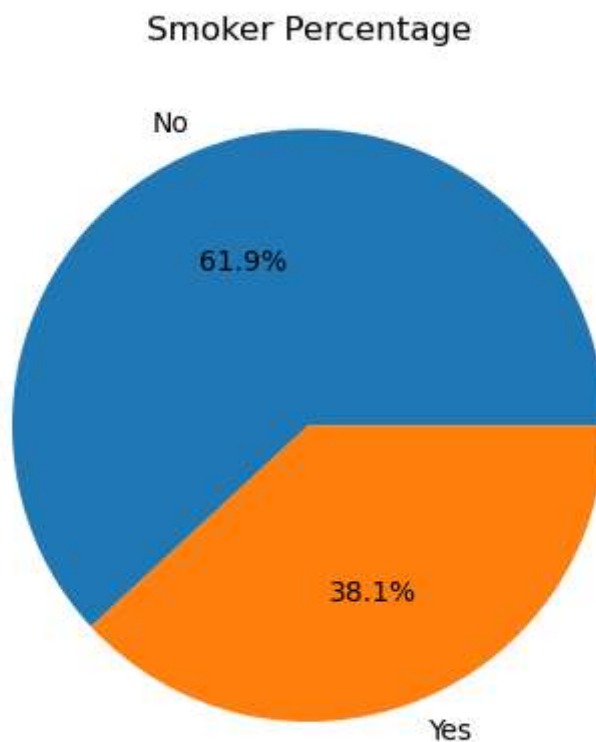
```
In [33]: df = sns.load_dataset('tips')  
plt.figure(figsize=(8,4))  
sns.boxplot(x=df['size'])  
plt.title("Box Plot of Size")  
plt.show()
```



```
In [34]: sns.countplot(x='day', data=df)
plt.title("Customer Count by Day")
plt.show()
```



```
In [35]: df['smoker'].value_counts().plot(
kind='pie',
autopct='%1.1f%%'
)
plt.title("Smoker Percentage")
plt.ylabel("")
plt.show()
```

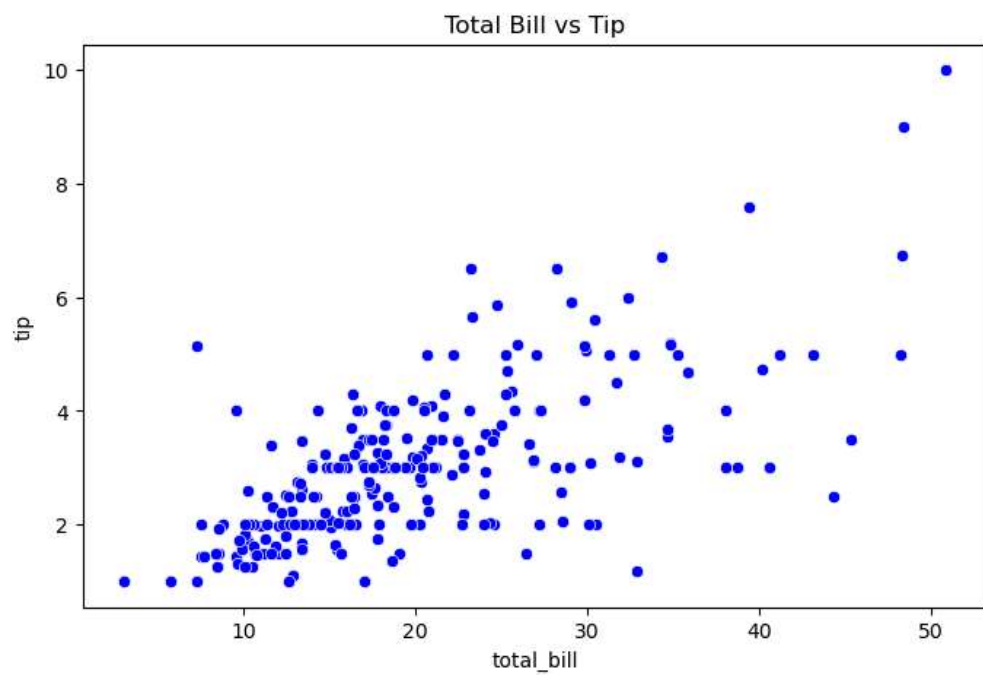


```
In [36]: import pandas as pd
import seaborn as sns

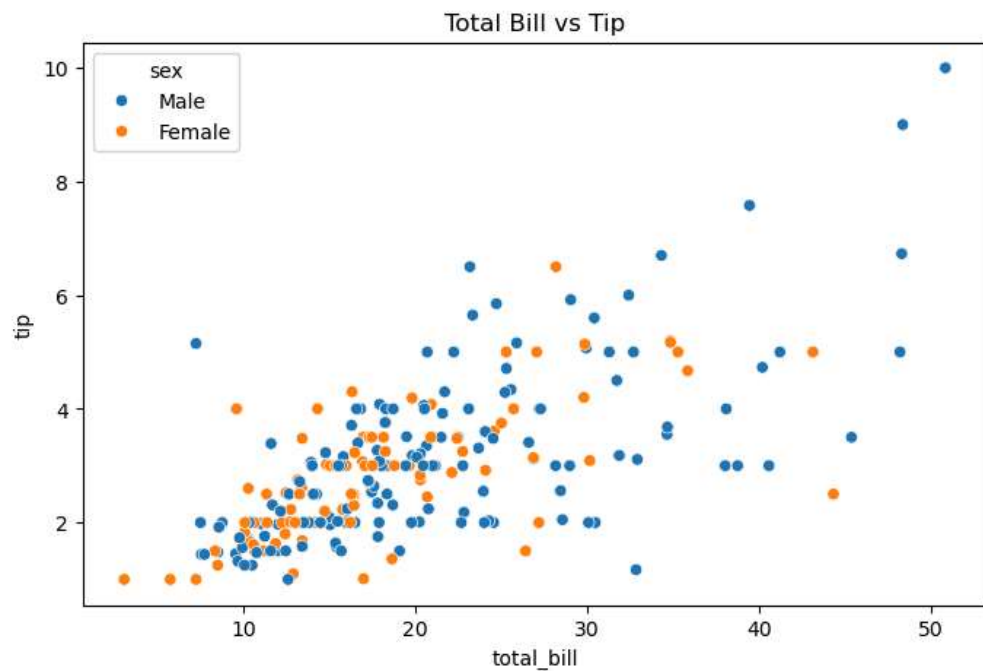
df = sns.load_dataset('titanic')
Q1 = df['fare'].quantile(0.25)
Q3 = df['fare'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
outlier = df[(df['fare'] < lower) | (df['fare'] > (Q3 + 1.5 * IQR))]
print(f"Total outliers: 9")
```

Total outliers: 9

```
In [39]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset('tips')
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='total_bill',
    y='tip',
    color='blue',
    data=df
)
plt.title("Total Bill vs Tip")
plt.show()
```



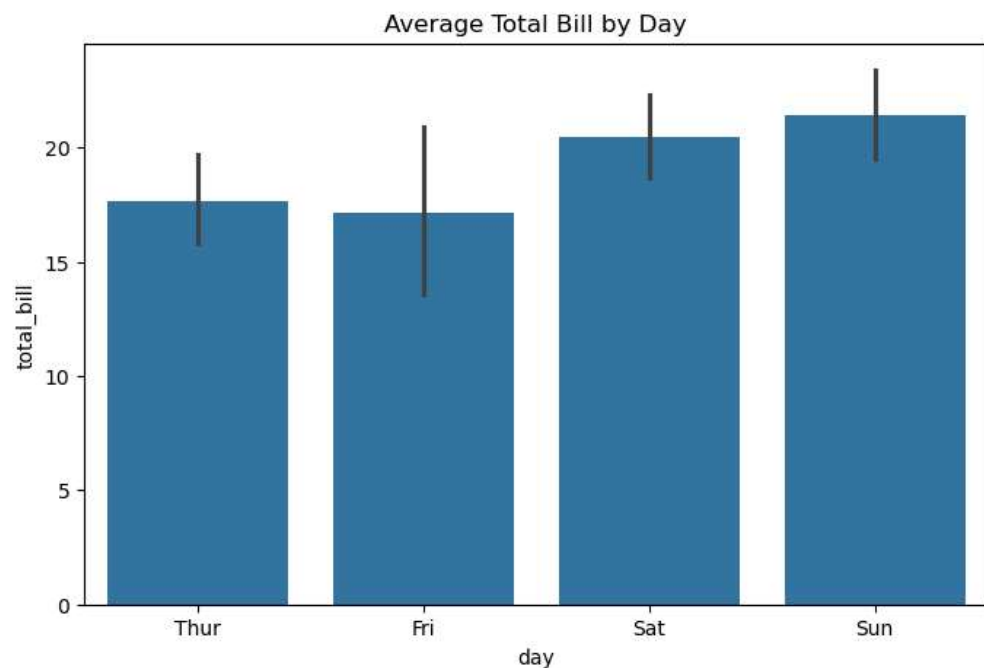
```
In [40]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset('tips')
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='total_bill',
    y='tip',
    hue='sex',
    data=df
)
plt.title("Total Bill vs Tip")
plt.show()
```



```
In [41]: corr = df['total_bill'].corr(df['tip'])
print(corr)
```

0.6757341092113641

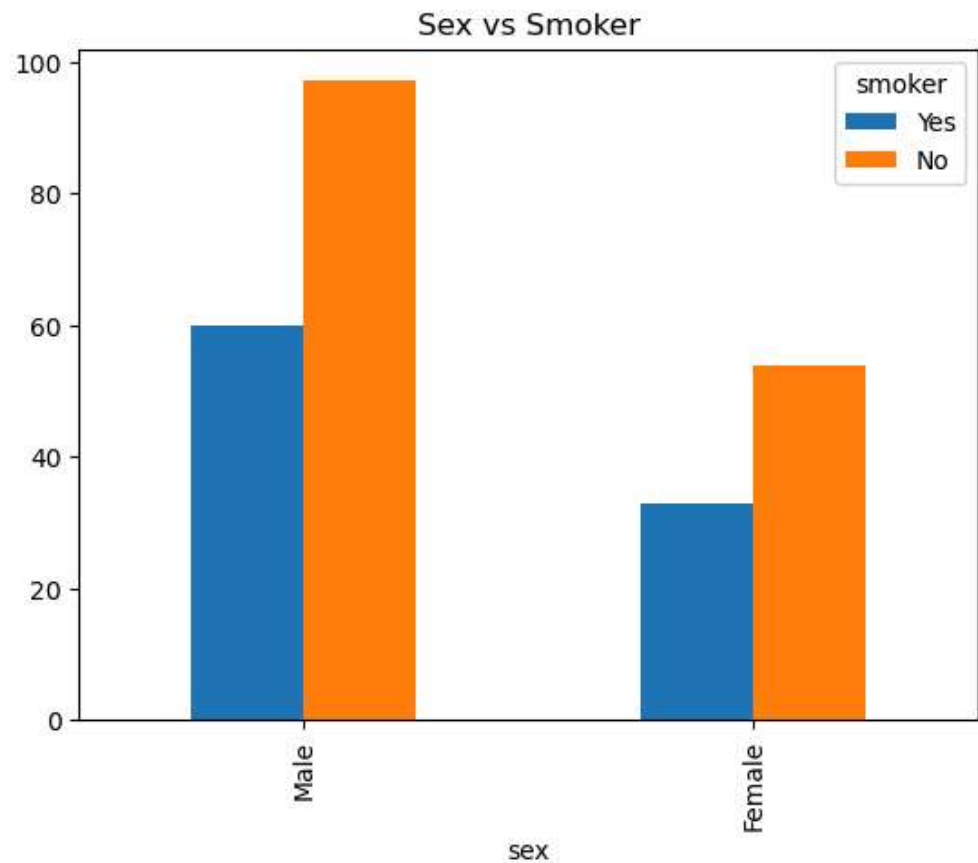
```
In [42]: plt.figure(figsize=(8,5))
sns.barplot(
x='day',
y= 'total_bill',
data=df
)
plt.title("Average Total Bill by Day")
plt.show()
```



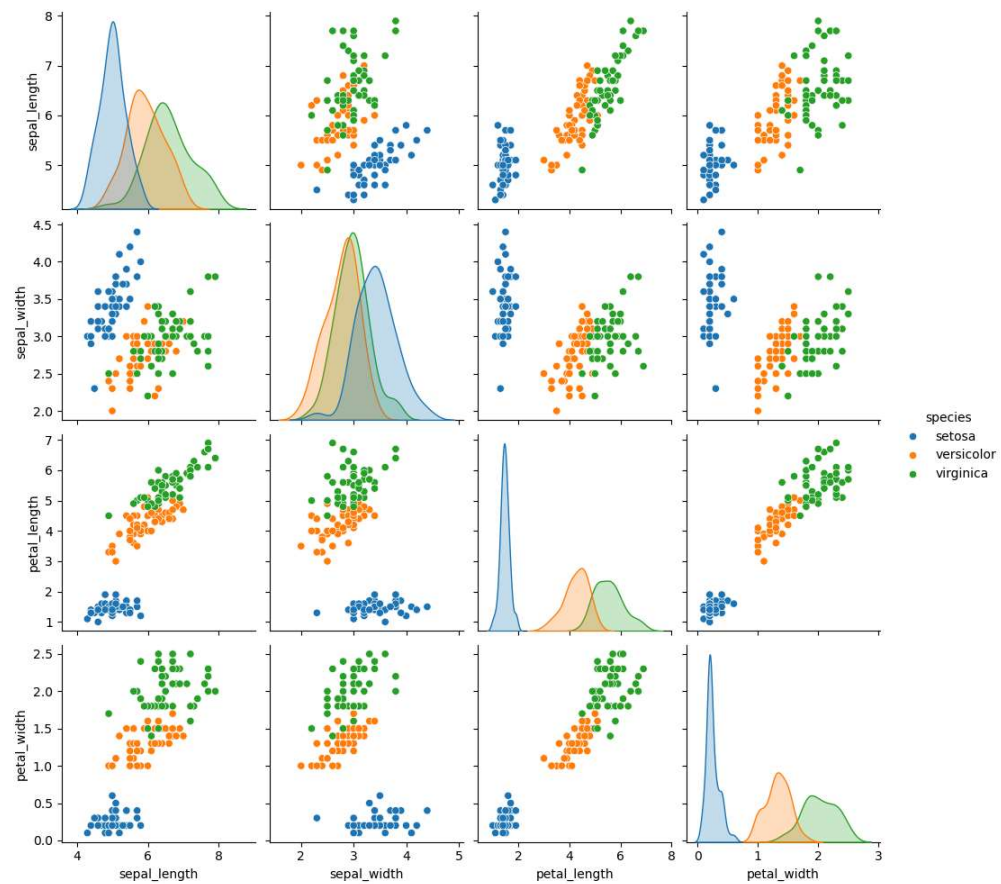
```
In [45]: print(
df.groupby('day', observed=False) ['total_bill'].mean()
)
print(
df.groupby('day', observed=False) ['total_bill'] .agg(['mean', 'median',
'min', 'max'])
)
```

```
day
Thur    17.682742
Fri     17.151579
Sat     20.441379
Sun     21.410000
Name: total_bill, dtype: float64
      mean  median  min  max
day
Thur  17.682742   16.20  7.51  43.11
Fri   17.151579   15.38  5.75  40.17
Sat   20.441379   18.24  3.07  50.81
Sun   21.410000   19.63  7.25  48.17
```

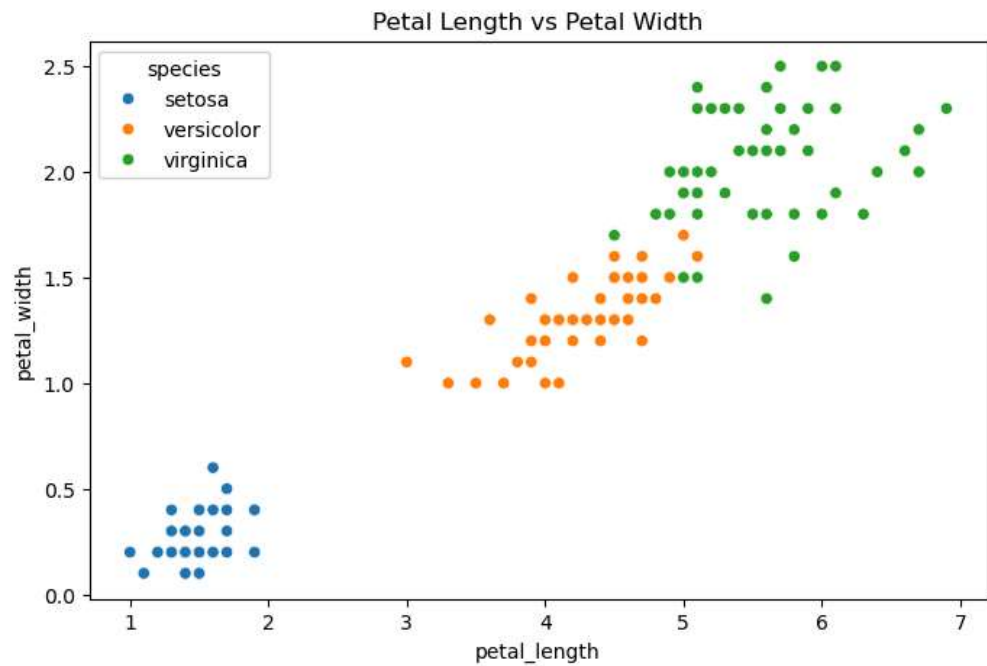
```
In [46]: pd.crosstab(  
df['sex'],  
df['smoker']  
) .plot(kind='bar')  
plt.title("Sex vs Smoker")  
plt.show()
```



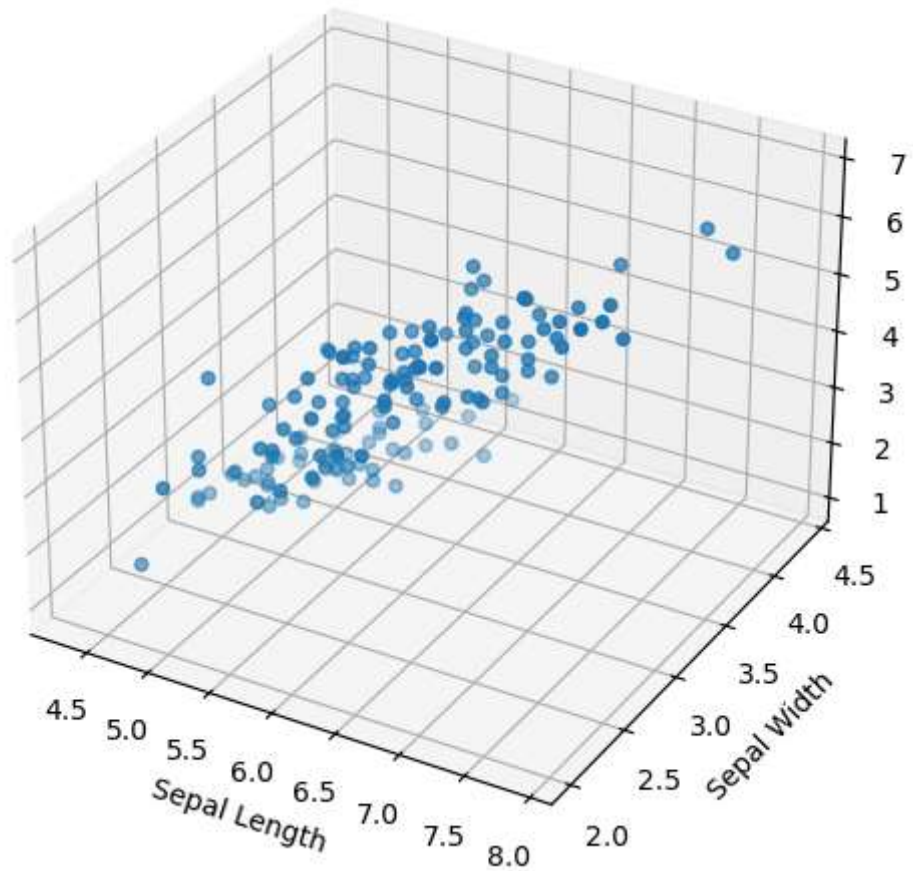
```
In [47]: import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset("iris")
sns.pairplot(df,
             hue='species'
             )
plt.show()
```



```
In [48]: plt.figure(figsize=(8,5))
sns.scatterplot(
x= 'petal_length',
y='petal_width',
hue='species',
data=df
)
plt.title("Petal Length vs Petal Width")
plt.show()
```



```
In [49]: from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(
df['sepal_length'],
df['sepal_width'],
df['petal_length'])
ax.set_xlabel('Sepal Length')
ax.set_ylabel('Sepal Width')
ax.set_zlabel('Petal Length')
plt.show()
```



```

In [54]: import pandas as pd
import numpy as np
import json
import re
df = pd.read_csv('walmart-products.csv')
df['price'] = df['final_price'].apply(lambda x:
float(str(x).replace('$','')))
def get_val(specs, key):
    try:
        return next((i['value'] for i in json.loads(specs) if
i['name']==key), np.nan)
    except:
        return np.nan
df['brand'] = df['brand'].fillna(df['specifications'].apply(lambda x:
get_val(x, 'Brand')))
df['color'] = df['specifications'].apply(lambda x: get_val(x, 'Color'))
df['material'] = df['specifications'].apply(lambda x: get_val(x,
'Material'))
df['rating'] = pd.to_numeric(df['rating_stars'], errors='coerce')
df['reviews'] = df['review_count'].fillna(0).astype(int)
for col in ['available_for_delivery', 'available_for_pickup',
'free_returns']:
    df[col] = df[col].astype(str).str.lower().isin(['true','1'])
df['category'] = df['category_path'].fillna('').apply(lambda x:
x.split('/')[1] if len(x.split('/'))>1 else 'Other')
clean = df[['product_id','product_name','brand','price','rating','reviews',
'available_for_delivery','color','material','category']].copy()
clean.columns =
['id','name','brand','price','rating','reviews','delivery','color','material',
'category']
clean.to_csv('walmart_clean.csv', index=False)
print("✅ Cleaned CSV saved as 'walmart_clean.csv'")
print(f"📊 Shape: {clean.shape}")
print(f"📄 Preview:\n{clean.head()}")

```

✅ Cleaned CSV saved as 'walmart_clean.csv'

📊 Shape: (1000, 10)

📄 Preview:

	id	name	brand	price	rating	reviews	delivery	color	
0	173530386	Laura Mercier Caviar Stick Eye Color Sugar Fro...							
1	430528189	Exultantex Grey Blackout Curtains for Living R...							
2	6013308220	Jessica London Women's Plus Size Stretch Knit ...							
3	161657830	100% Cotton King Percale Duvet Set (3 Piece) B...							
4	5397071399	Disney Boys Graphic Tee Donald Duck Short Slee...							

	brand	price	rating	reviews	delivery	color	
0	Laura Mercier	22.90	NaN	6	True	NaN	
1	Exultantex	47.88	NaN	58	True	Gray	
2	Jessica London	33.24	NaN	2	True	NaN	
3	Simply Put	49.99	NaN	101	True	Blue	
4	Disney	12.99	NaN	15	True	NaN	

	material	category
0	NaN	cp
1	Triple Weave Thermal Fabric	cp
2	NaN	cp
3	Cotton	cp
4	67% Polyester, 5% Spandex, 28% Acetate	cp

```

In [4]: import pandas as pd
import numpy as np
import json
import re
df = pd.read_csv('walmart-products.csv')
df['price'] = df['final_price'].apply(lambda x:
float(str(x).replace('$','')))
def get_val(specs, key):
    try:
        return next((i['value'] for i in json.loads(specs) if
i['name']==key), np.nan)
    except:
        return np.nan
df['brand'] = df['brand'].fillna(df['specifications'].apply(lambda x:
get_val(x, 'Brand')))
df['color'] = df['specifications'].apply(lambda x: get_val(x, 'Color'))
df['material'] = df['specifications'].apply(lambda x: get_val(x,
'Material'))
df['rating'] = pd.to_numeric(df['rating_stars'], errors='coerce')
df['reviews'] = df['review_count'].fillna(0).astype(int)
for col in ['available_for_delivery', 'available_for_pickup',
'free_returns']:
    df[col] = df[col].astype(str).str.lower().isin(['true','1'])
df['category'] = df['category_path'].fillna('').apply(lambda x:
x.split('/')[1] if len(x.split('/'))>1 else 'Other')
clean = df[['product_id','product_name','brand','price','rating','reviews',
'available_for_delivery','color','material','category']].copy()
clean.columns =
['id','name','brand','price','rating','reviews','delivery','color','material',
'category']
clean.to_csv('walmart_clean.csv', index=False)
print("Cleaned CSV saved as 'walmart_clean.csv'")
print(f" Shape: {clean.shape}")
print(f"\n Preview:\n{clean.head()}")

```

Cleaned CSV saved as 'walmart_clean.csv'

Shape: (1000, 10)

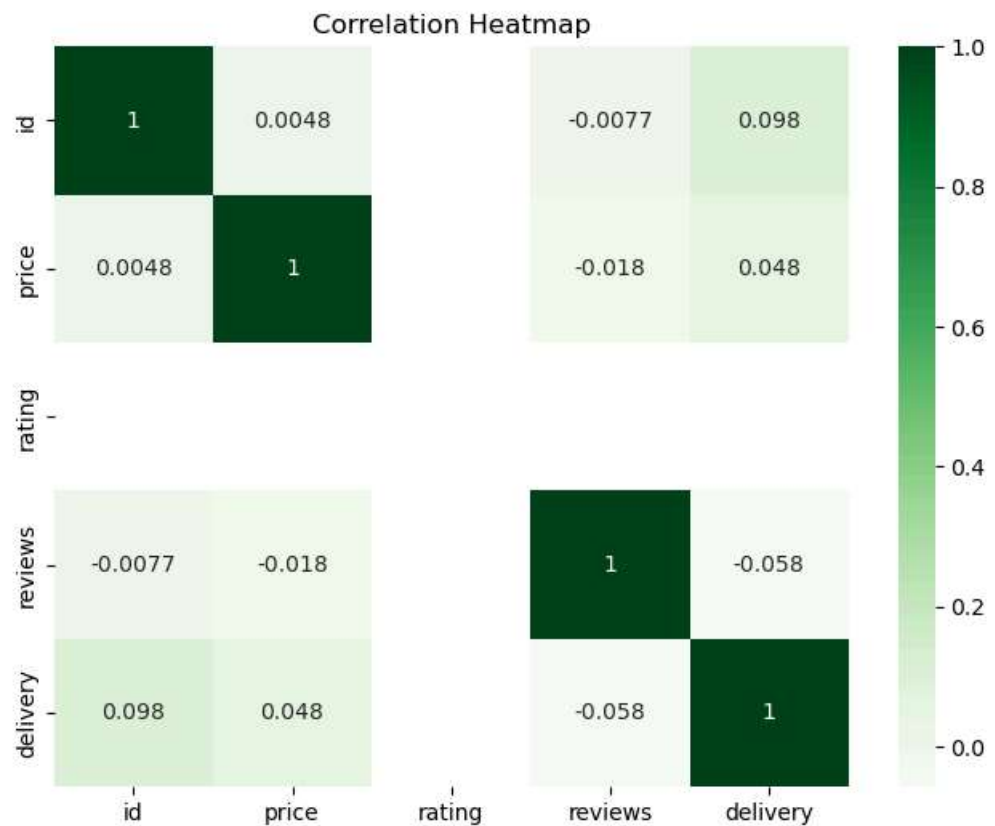
Preview:

	id	name \
0	173530386	Laura Mercier Caviar Stick Eye Color Sugar Fro...
1	430528189	Exultantex Grey Blackout Curtains for Living R...
2	6013308220	Jessica London Women's Plus Size Stretch Knit ...
3	161657830	100% Cotton King Percale Duvet Set (3 Piece) B...
4	5397071399	Disney Boys Graphic Tee Donald Duck Short Slee...

	brand	price	rating	reviews	delivery	color \
0	Laura Mercier	22.90	NaN	6	True	NaN
1	Exultantex	47.88	NaN	58	True	Gray
2	Jessica London	33.24	NaN	2	True	NaN
3	Simply Put	49.99	NaN	101	True	Blue
4	Disney	12.99	NaN	15	True	NaN

	material	category
0	NaN	cp
1	Triple Weave Thermal Fabric	cp
2	NaN	cp
3	Cotton	cp
4	67% Polyester, 5% Spandex, 28% Acetate	cp

```
In [11]: df = pd.read_csv('walmart_clean.csv')
plt.figure(figsize=(8, 6))
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='Greens'
)
plt.title("Correlation Heatmap")
plt.show()
```



```

In [19]: import pandas as pd
import numpy as np
from statsmodels.stats.outliers_influence import variance_inflation_factor
data = {
    'sepal_length': [5.1, 4.9, 4.7, 4.6, 5.0],
    'sepal_width': [3.5, 3.0, 3.2, 3.1, 3.6],
    'petal_length': [1.4, 1.4, 1.3, 1.5, 1.4],
    'petal_width': [0.2, 0.2, 0.2, 0.1, 0.2],
    'species': [0, 0, 0, 0, 0]
}
df = pd.DataFrame(data)

# Calculate VIF
X = df.drop("species", axis=1)
vif = pd.DataFrame()
vif["Feature"] = X.columns
vif["VIF"] = [
    variance_inflation_factor(X.values, i)
    for i in range(X.shape[1])
]
print("\nVIF Results:")
print(vif)

```

```

VIF Results:

```

	Feature	VIF
0	sepal_length	13063.547754
1	sepal_width	606.820731
2	petal_length	5461.684842
3	petal_width	585.898895

```

In [30]: import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap
(X_new.corr(),
 annot=True,
 cmap="coolwarm"
)
plt.title("After VIF Treatment")
plt.show()

```

Cell In[30], line 6

```

    annot=True,
    ^

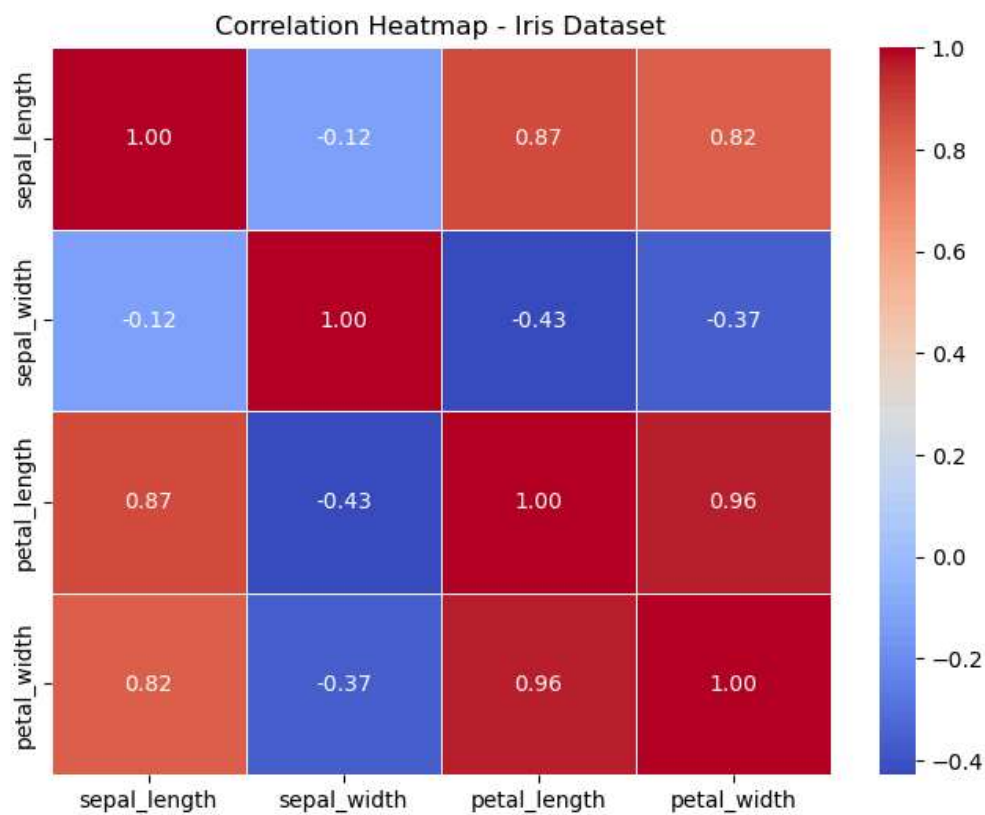
```

SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?

```
In [31]: import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('iris')
df_numeric = df.drop('species', axis=1)

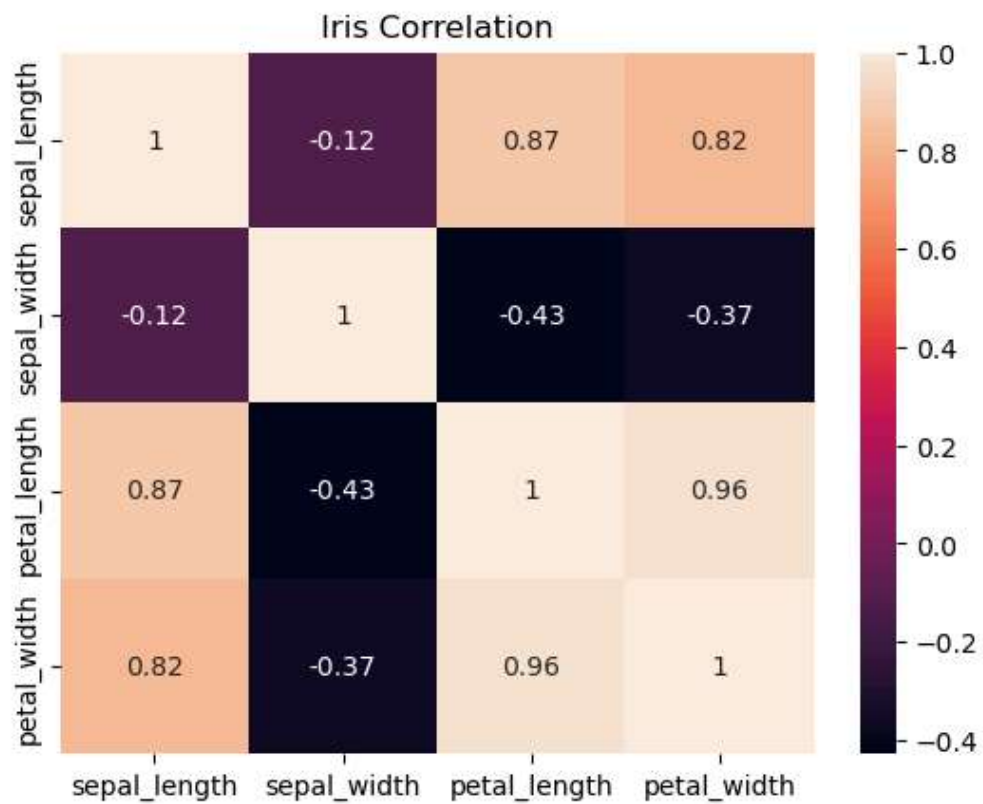
plt.figure(figsize=(8, 6))
sns.heatmap(
    df_numeric.corr(),
    annot=True,
    cmap='coolwarm',
    fmt='.2f',
    linewidths=0.5
)
plt.title('Correlation Heatmap - Iris Dataset')
plt.show()
```



```
In [32]: import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('iris')

sns.heatmap(df.drop('species', axis=1).corr(), annot=True)
plt.title('Iris Correlation')
plt.show()
```



```
In [ ]:
```